

ROADASSIST BHARAT

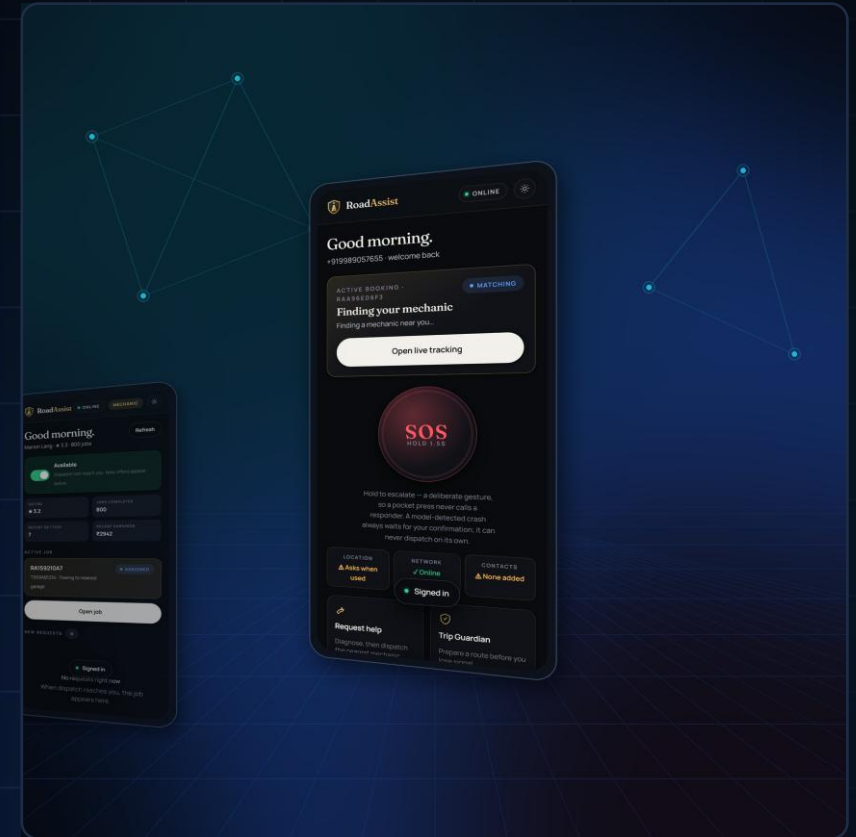
AI-POWERED ROADSIDE ASSISTANCE
& EMERGENCY MOBILITY PLATFORM

AI

CLOUD

MOBILITY

SAFETY



A working platform used to demonstrate Cloud Computing Modules 1-4

THE REAL-WORLD PROBLEM

WHEN THE ROAD STOPS, THE PROBLEM STARTS.



Every one of these six failures is an information problem before it is a mechanical one.

The driver cannot diagnose, cannot locate, cannot summon, and cannot see what happens next.

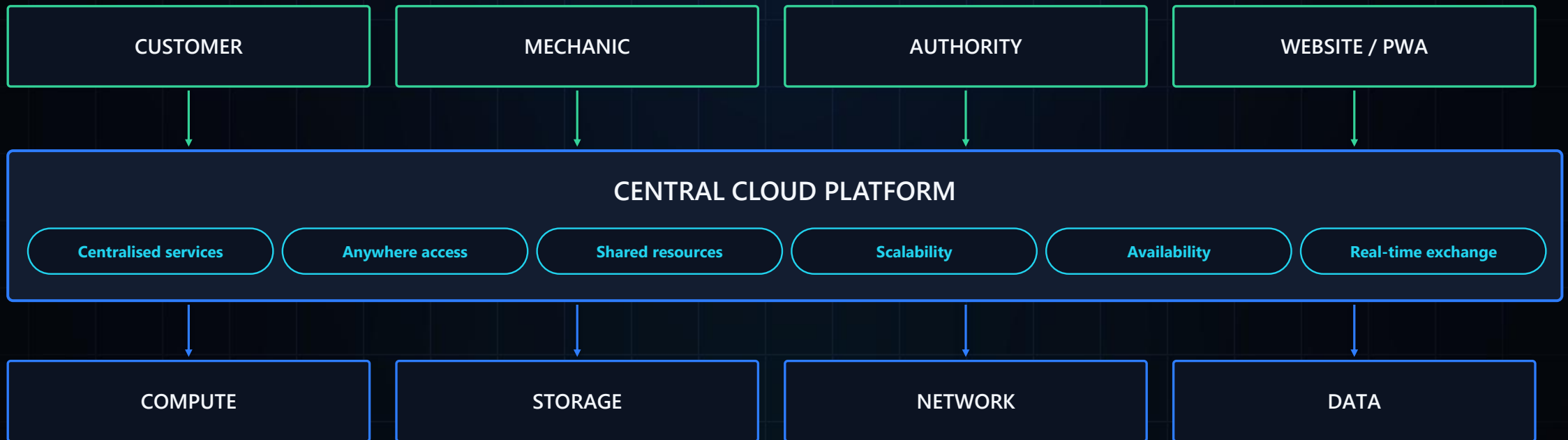
A cloud platform is the natural answer, because the information needed - who is nearby, what the fault is, where the vehicle is, who has been told - lives in different places and must be brought together in real time.

ONE PLATFORM. EVERY ROAD. EVERY RESPONSE.



The cloud platform is the backbone: the only component every actor talks to, and the only place that knows the whole truth about a booking, an incident or a vehicle.

WHY ROADASSIST NEEDS THE CLOUD



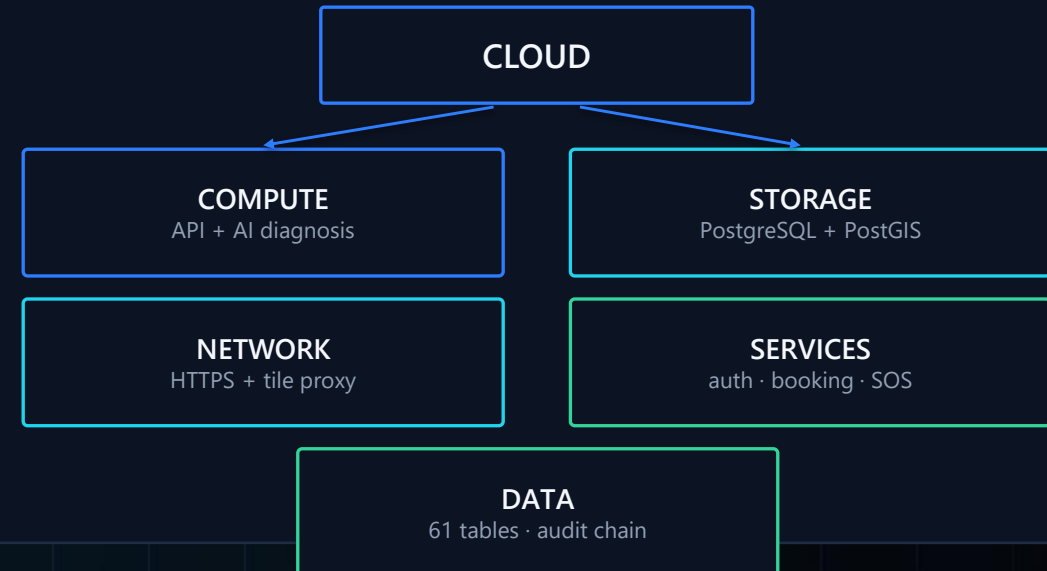
Without the cloud, each actor would hold its own copy of the truth and reconcile it later. With it, one authoritative service is reachable from any road, on any device, at any hour.

MODULE 1 — UNDERSTANDING CLOUD COMPUTING

THE FIVE ESSENTIAL CHARACTERISTICS

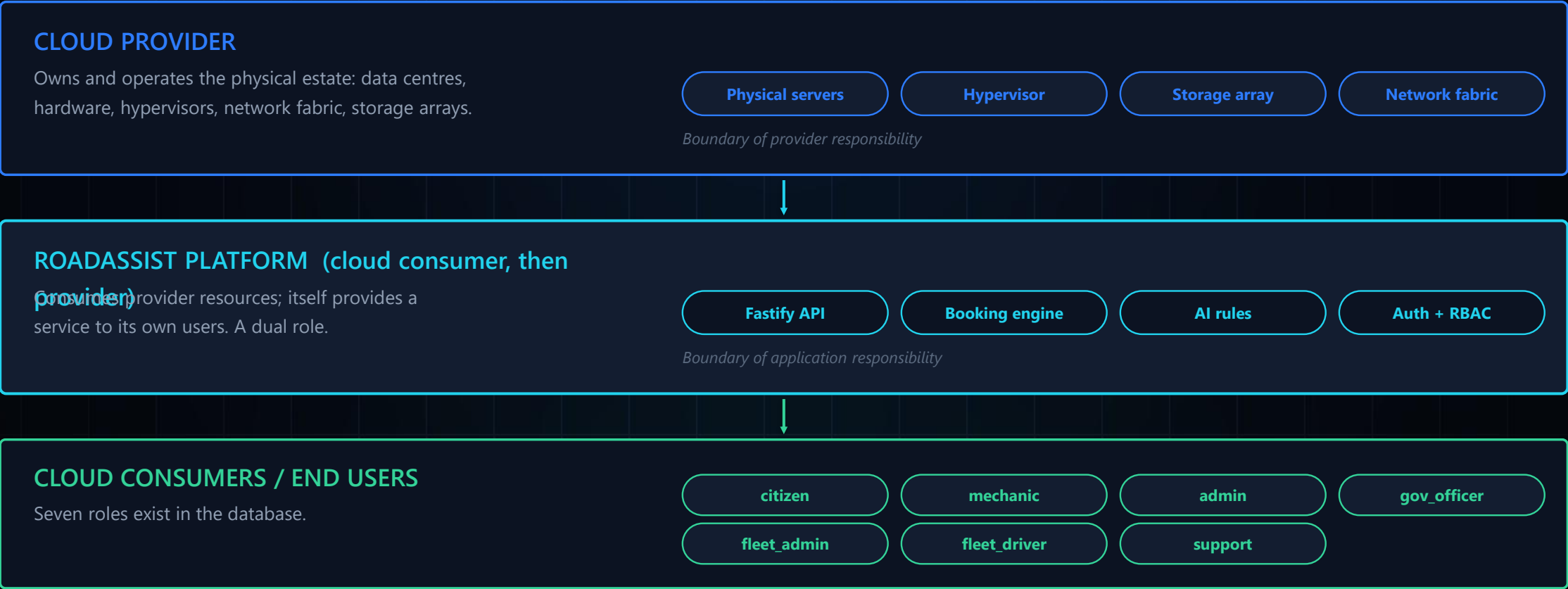
- **On-demand self-service**
A user signs in and books assistance; nobody provisions anything.
- **Broad network access**
Reachable from any phone browser - the PWA needs no install.
- **Resource pooling**
One platform, one database, one API serve every customer and mechanic.
- **Rapid elasticity**
Capacity can follow demand; festival and monsoon peaks are not steady state.
- **Measured service**
Usage is observable per request, which is what makes scaling decidable.

WHAT THE CLOUD SUPPLIES ROADASSIST



Cloud computing grew out of utility computing, grid computing and virtualisation. RoadAssist consumes it the way a home consumes electricity: on demand, metered, without owning the plant.

WHO USES THE CLOUD?



WHERE DOES ROADASSIST FIT?

Architectural mapping. The project runs on local containers in development; no commercial cloud provider is used.

MODEL	WHO MANAGES WHAT	ROADASSIST MAPPING	STATUS
IaaS	Provider gives virtual machines, storage and network; you manage the OS up	Where the containers and database would run in a deployed environment	MAPPING
PaaS	Provider also manages runtime and scaling; you deploy code only	Fastify API could be deployed as a managed Node service	MAPPING
SaaS	Consumer uses finished software over the network	What the customer, mechanic and authority actually experience	IMPLEMENTED

DEPLOYMENT MODELS

PUBLIC Shared provider estate. Lowest cost, least control.	PRIVATE Dedicated estate. Suits the government / RAKSHA tenant.	HYBRID Public for the citizen app, private for authority data.
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CLOUD BENEFITS VS CLOUD CHALLENGES

BENEFITS

Scalability

One platform serves a city or a state without redesign.

Availability

A breakdown at 2am must reach a live service.

Centralised services

One authoritative booking state, not many copies.

Remote access

Any phone browser, no install, no app store.

Resource sharing

Customers, mechanics and authorities share one estate.

Operational flexibility

Providers are swappable behind adapters.

RISKS AND CHALLENGES

Security

Handled: OTP, rotating refresh tokens, resource-scoped RBAC.

Privacy

Medical data is break-glass only, and every read is logged.

Network dependency

Answered: offline-first PWA with an idempotent sync journal.

Availability dependency

An emergency path cannot fail silently - SOS never queues.

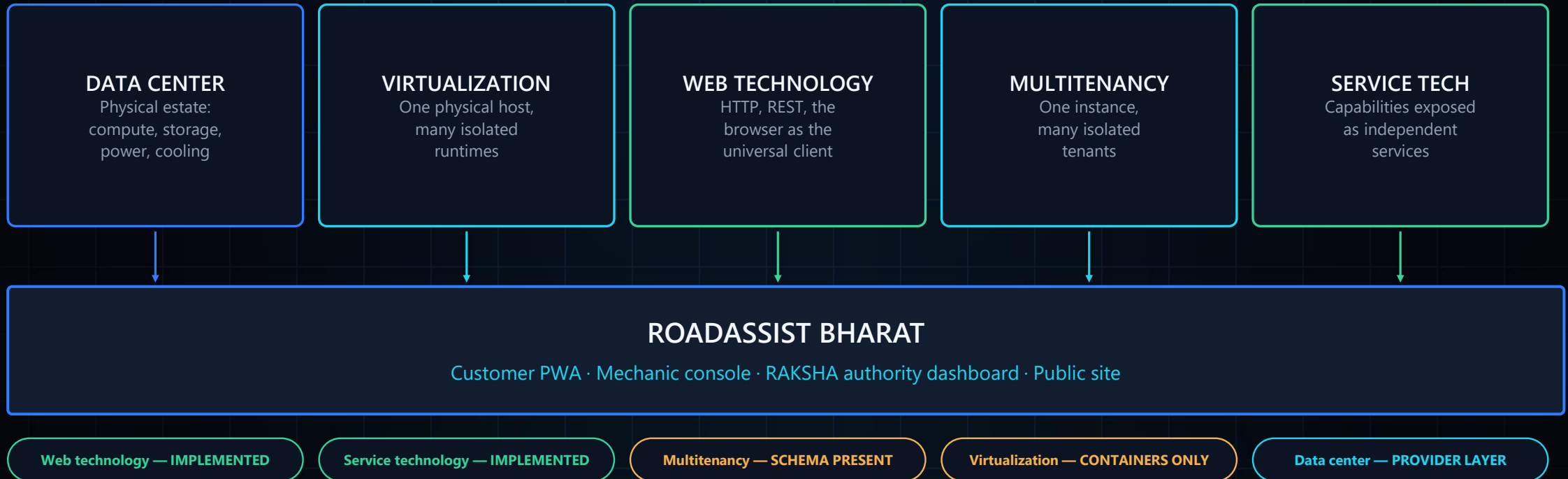
Resource management

Rate limits per number and per IP guard shared capacity.

Data protection

Hash-chained audit log makes tampering detectable.

MODULE 2 — CLOUD ENABLING TECHNOLOGY



Honest position: the top two are built into the application; multitenancy exists in the schema; virtualization is container-level, not a hypervisor we operate; the data centre belongs to whichever provider hosts it.

THE CLOUD NEEDS A PHYSICAL FOUNDATION.



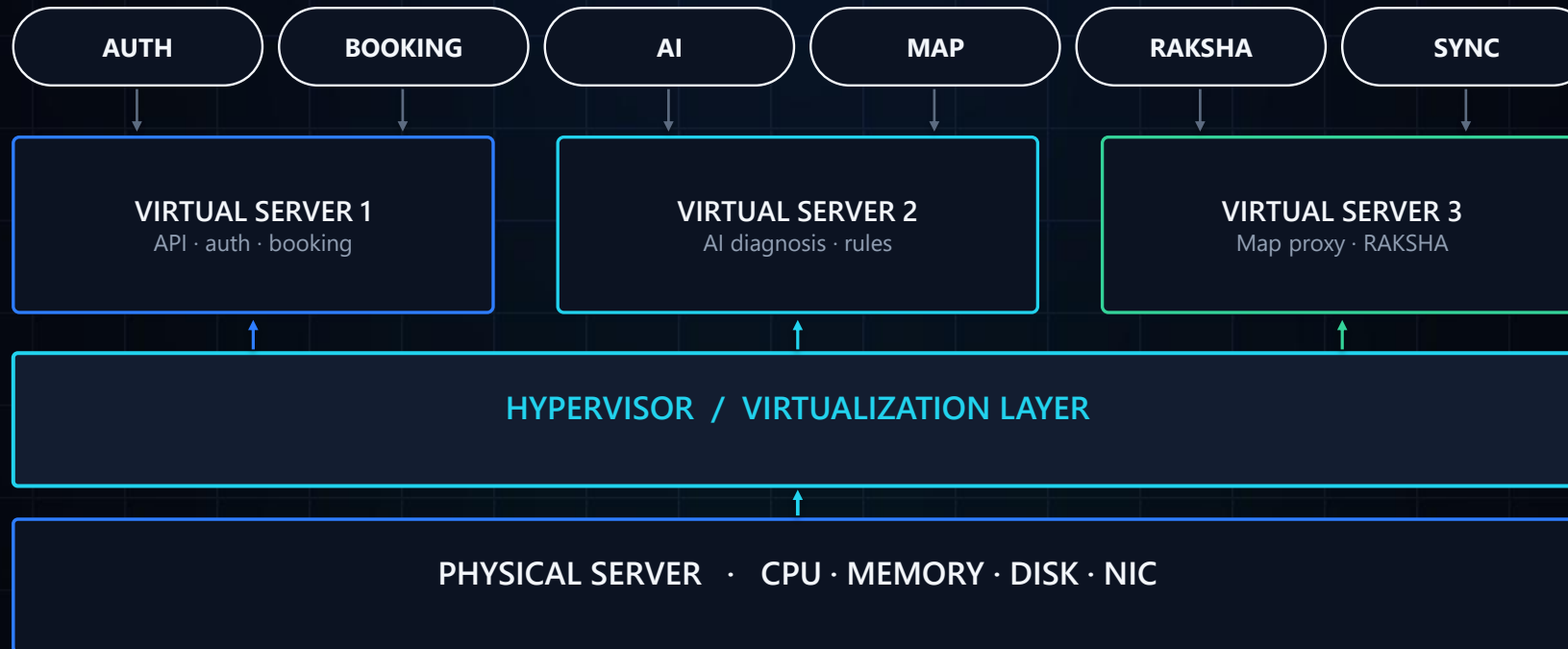
WHAT A DATA CENTRE PROVIDES

- Compute**
Racked servers running virtualized workloads
- Storage**
Disk arrays and storage networks
- Networking**
Switching fabric, routing, external links
- Power**
Redundant supply and battery / generator backup
- Cooling**
Thermal management that bounds density
- Monitoring**
Environmental and hardware telemetry



ONE PHYSICAL SYSTEM. MULTIPLE VIRTUAL RESOURCES.

Honest scope: RoadAssist uses OS-level virtualization (Docker containers) in development. We do not operate a hypervisor — that layer belongs to the infrastructure provider.



FROM CLOUD INFRASTRUCTURE TO USER SERVICES

WEB TECHNOLOGY

IMPLEMENTED

HTTP / REST

58 JSON routes, uniform error envelope

Browser client

No install; the URL is the app

PWA

Manifest + service worker, installable

Offline shell

19 assets cached for a dead network

MULTITENANT TECHNOLOGY

SCHEMA PRESENT

One instance

A single deployment serves every tenant

Fleet tenancy

fleets + fleet_members tables exist

Gov tenancy

gov_jurisdictions + gov_officers

Isolation by role

Queries scoped to the caller, not the route

SERVICE TECHNOLOGY

IMPLEMENTED

AUTH / OTP

BOOKING

AI DIAGNOSIS

MAP

RAKSHA / SOS

PAYMENT

OFFLINE SYNC

AUDIT

Each capability is an independently addressable service behind one API contract.

CONSUMING A CLOUD SERVICE: PAYMENTS

ROADASSIST — cloud consumer

1 Invoice created

Amount is the invoice total.
Never read from the request.

2 Create order

Server-to-server, HTTP Basic.
POST /v1/orders

3 Checkout opens

Client gets the key ID only.
The key SECRET never leaves us.

4 Signature verified

HMAC-SHA256(order|payment).
No match, no settlement.

RAZORPAY — cloud provider (SaaS)

Default provider is mock. Live keys need a KYC-verified account.

TWO SETTLEMENT PATHS — ON PURPOSE

BROWSER CALLBACK

POST /v1/payments/id/confirm — the client returns the signed payload

WEBHOOK · the one that matters

POST /v1/webhooks/razorpay — signature over the raw body IS the auth

WHY A WEBHOOK AT ALL?

A customer can pay and immediately close the tab, lose signal, or have the page killed. The money has still moved. The browser is not a reliable narrator, so settlement cannot depend on it.

VERIFIED — 22 AUTOMATED CHECKS

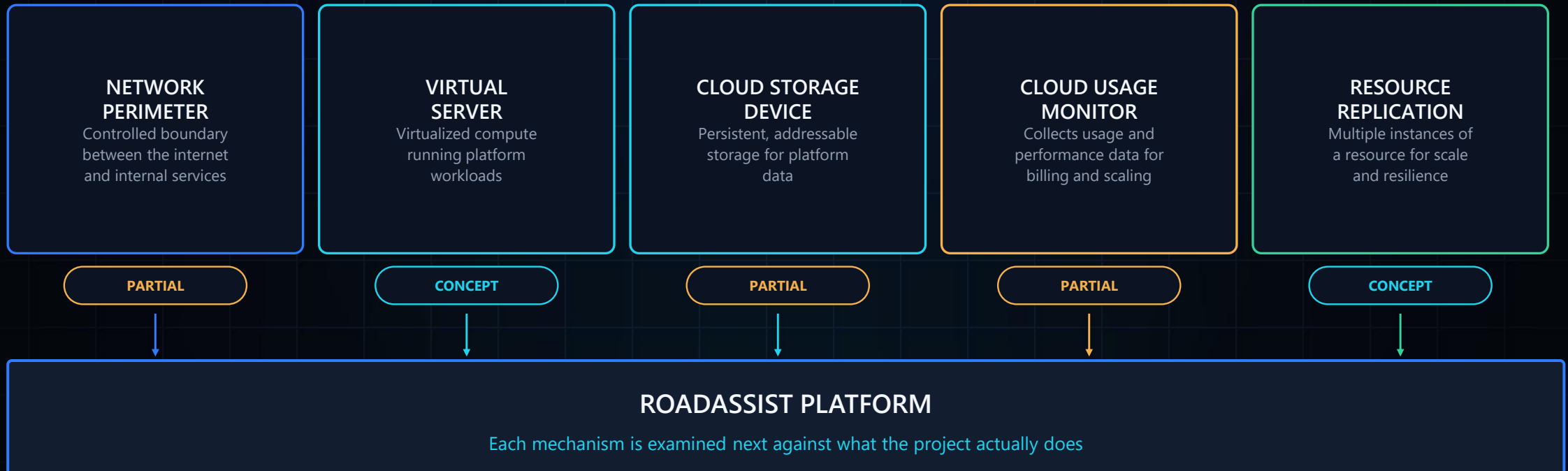
Forged signature refused

Unsigned webhook 401

Amount mismatch refused

Replay settles once

MODULE 3 — CLOUD INFRASTRUCTURE MECHANISMS



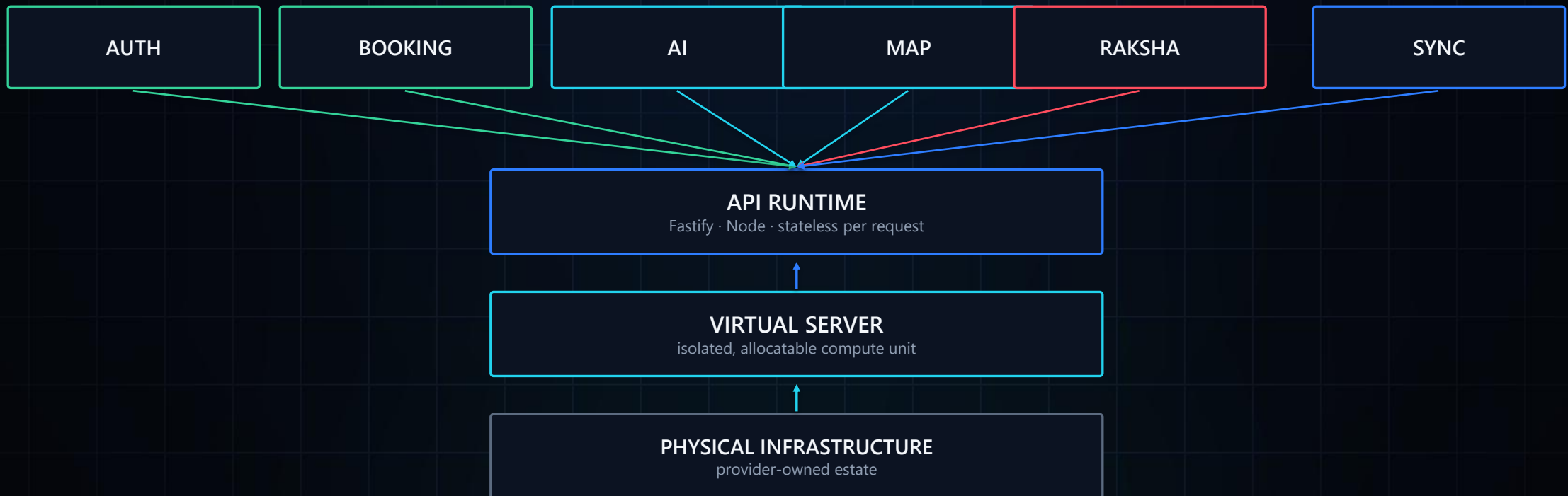
PARTIAL means a real but limited form exists — for example a health endpoint reporting database latency is a usage monitor, but not a full monitoring stack. CONCEPT means the mechanism is designed for, not built.

PROTECTING THE CLOUD BOUNDARY



Verified: a forged X-Forwarded-For header from an untrusted peer is ignored, so the per-IP rate limit cannot be bypassed. No specific firewall product or cloud provider service is claimed.

VIRTUAL COMPUTE FOR ROADASSIST SERVICES



Architectural concept. Today the API runs as a single Node process; the isolation boundary shown is the deployment target, not current infrastructure.

DATA MUST BE AVAILABLE WHEN USERS NEED IT.

CLOUD STORAGE — WHAT WE HOLD

IMPLEMENTED

Identity & sessions

users, roles, sessions, otp_challenges

Vehicle data

vehicles, user_vehicles, documents

Booking records

bookings, booking_events, offers

Service & money

invoices, payments, reviews

Emergency

incidents, responses, emergency_contacts

RAKSHA operations

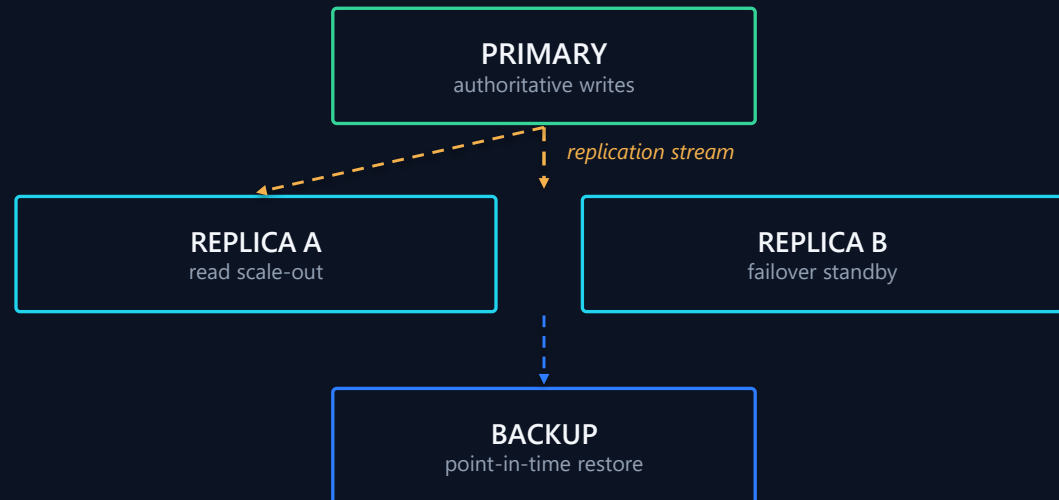
edge_devices, detections, road_segments

Audit

hash-chained, append-only, tamper-evident

RESOURCE REPLICATION

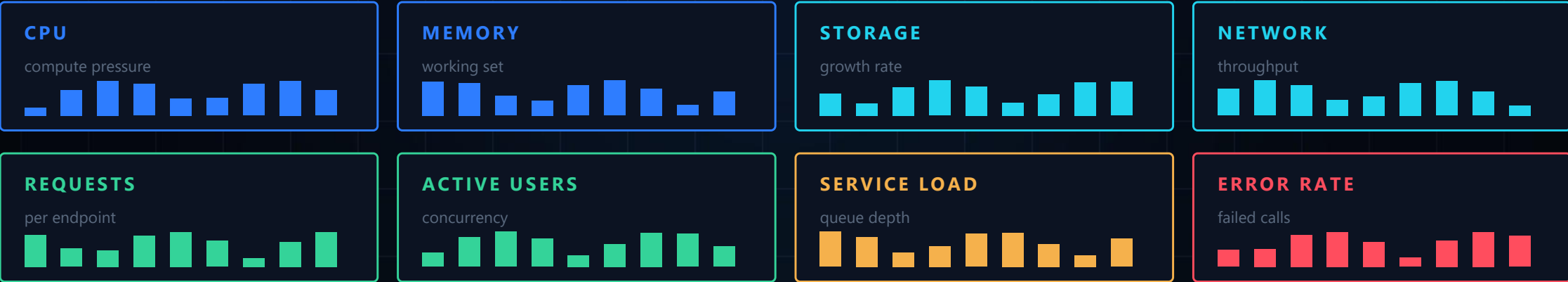
ARCHITECTURAL CONCEPT



Today: a single PostgreSQL instance. No replica is configured — stated, not implied.

OBSERVE. MEASURE. OPTIMIZE.

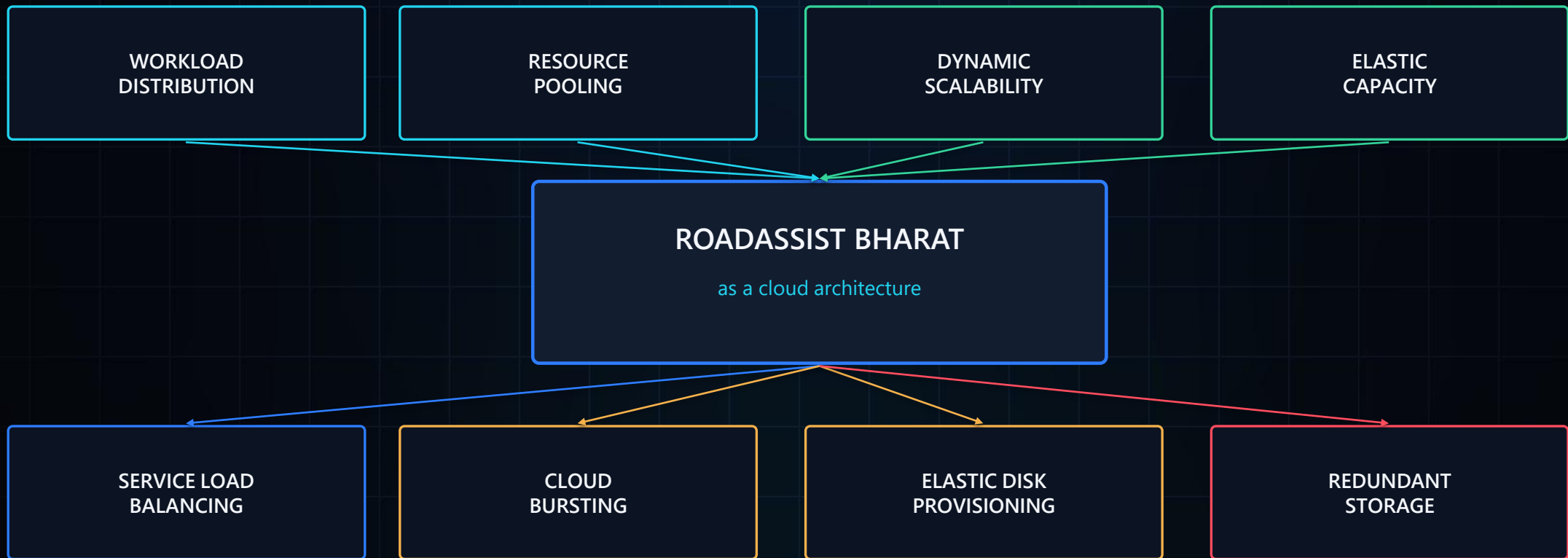
Cloud usage monitoring architecture. Metric values below are illustrative of the mechanism — no measurements are claimed.



WHAT ROADASSIST ACTUALLY MEASURES TODAY
GET /health returns database latency and live provider status · every request is logged with a unique id and response time · model_predictions records AI latency and fallback use



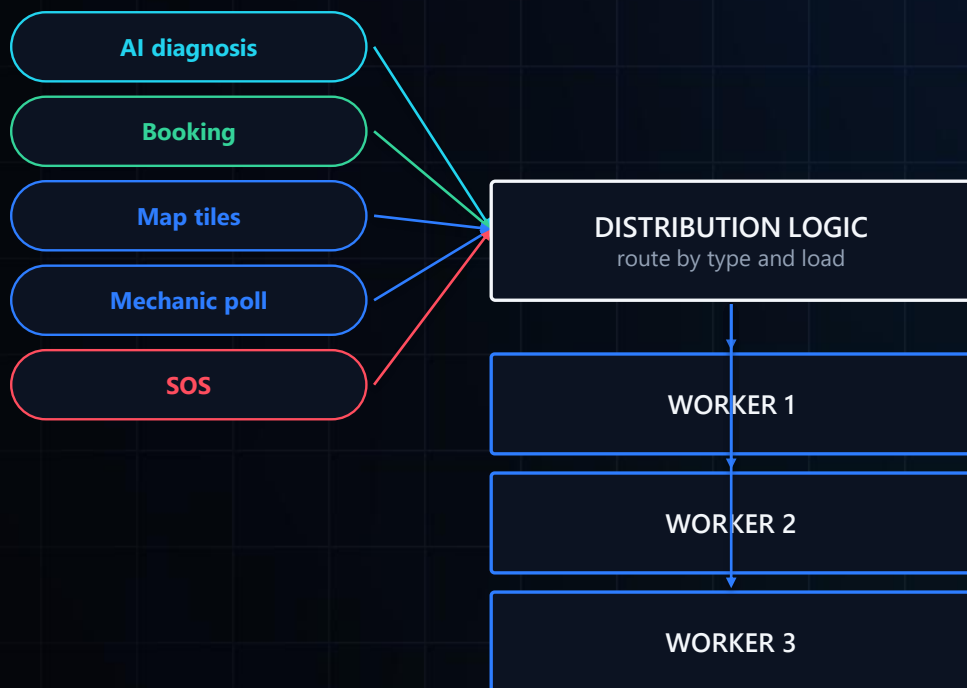
MODULE 4 — FUNDAMENTAL CLOUD ARCHITECTURES



Plus cloud operations: migration, static scheduling and dynamic scheduling — covered on slide 23.

DISTRIBUTE WORK. SHARE RESOURCES.

WORKLOAD DISTRIBUTION



RESOURCE POOLING



One pool serves every tenant and every request type. A map-tile burst and an AI diagnosis draw on the same underlying capacity, which is exactly why pooling improves utilisation over dedicated allocation.

Implemented today in a limited form: the map tile proxy keeps a shared in-memory tile cache, so one user's fetch serves the next user's request.

THE CLOUD ADAPTS TO DEMAND.



DYNAMIC SCALABILITY

Instances are added and removed automatically as measured demand crosses thresholds.

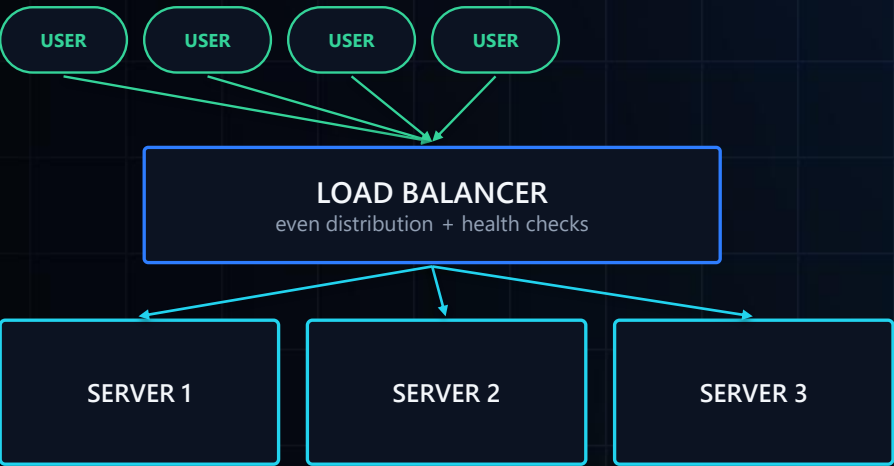
ELASTIC RESOURCE CAPACITY

Existing instances are resized — more CPU and memory allocated to the same server.

Architectural capability. RoadAssist does NOT implement automatic scaling. It is written to accept it: request handling is stateless, so instances are interchangeable.

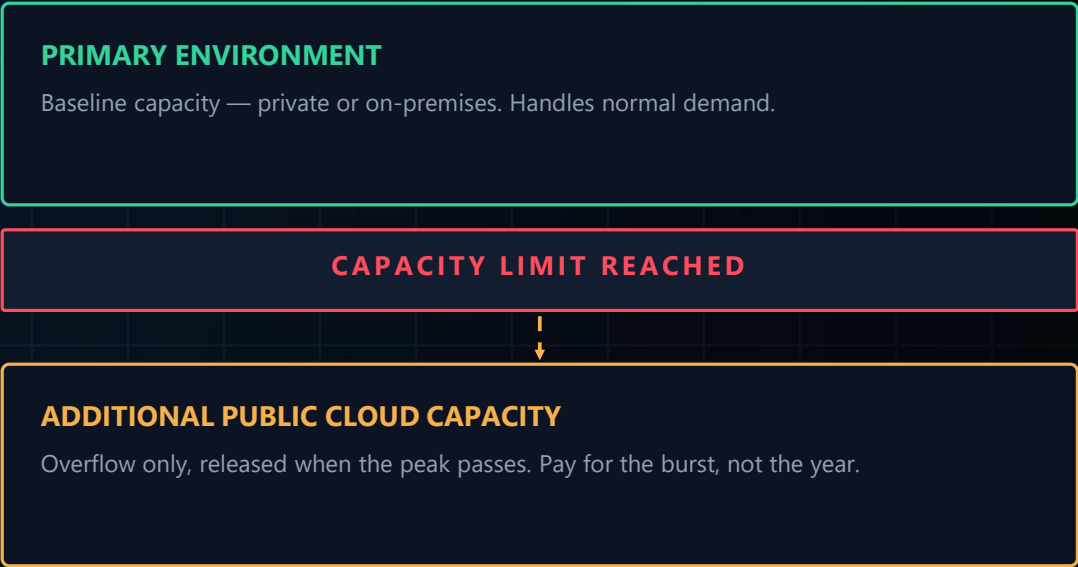
KEEPING SERVICES RESPONSIVE UNDER LOAD

SERVICE LOAD BALANCING



The balancer also removes an unhealthy instance from rotation, which is how load balancing contributes to availability, not only to throughput.

CLOUD BURSTING

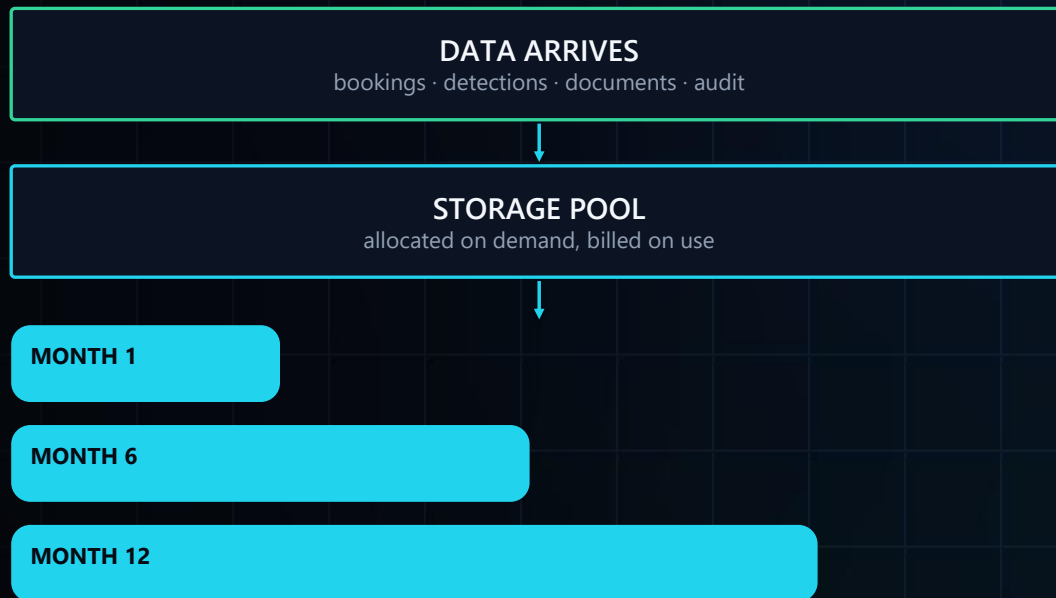


Natural fit: a festival weekend or a regional weather event is exactly a short, predictable burst.

Neither is currently implemented. RoadAssist runs a single API process behind an HTTPS tunnel for demonstration; both are described as target architecture.

STORAGE THAT GROWS WITH THE SYSTEM

ELASTIC DISK PROVISIONING



Capacity follows actual growth rather than being bought up-front for a size nobody can predict.

REDUNDANT STORAGE



- **Vehicle information**
registrations, class, documents
- **Booking records**
state history, offers, invoices
- **Service records**
completed work, reviews, ratings
- **Audit information**
hash-chained, must survive intact
- **RAKSHA operational data**
detections, road health, devices

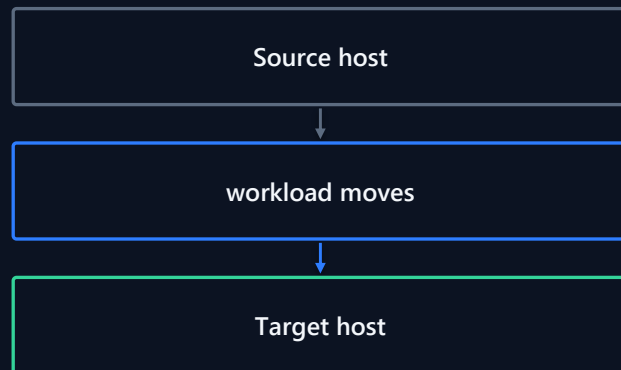
The audit chain is the strongest case: it is legally meaningful only if it cannot be lost.

Current implementation: one PostgreSQL instance with a fixed local volume. Both architectures on this slide are target state.

OPERATING THE CLOUD

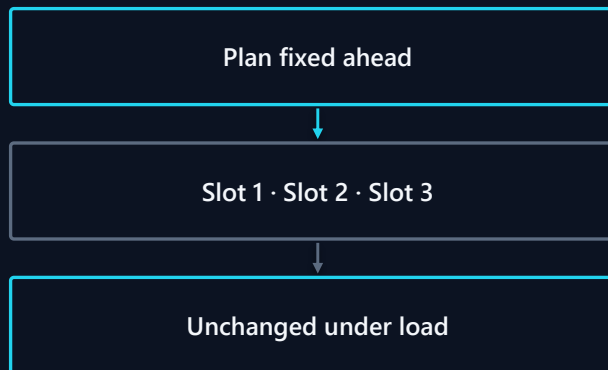
MIGRATION

Moving a workload or resource from one host to another — for maintenance, consolidation, cost, or to recover from a failing node.



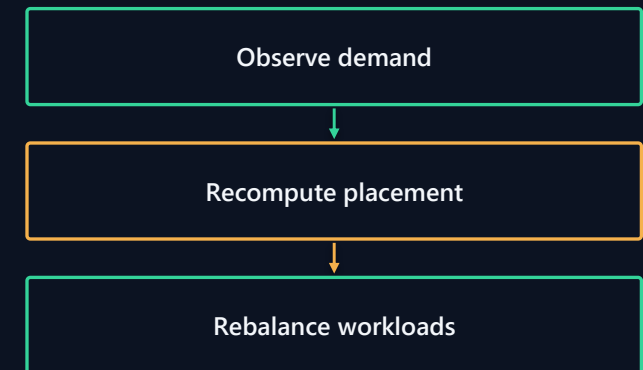
STATIC SCHEDULING

Allocation decided in advance and held. Predictable, simple to reason about, but blind to what actually happens at runtime.



DYNAMIC SCHEDULING

Allocation recomputed as demand changes. Higher utilisation, at the cost of needing live measurement to decide on.



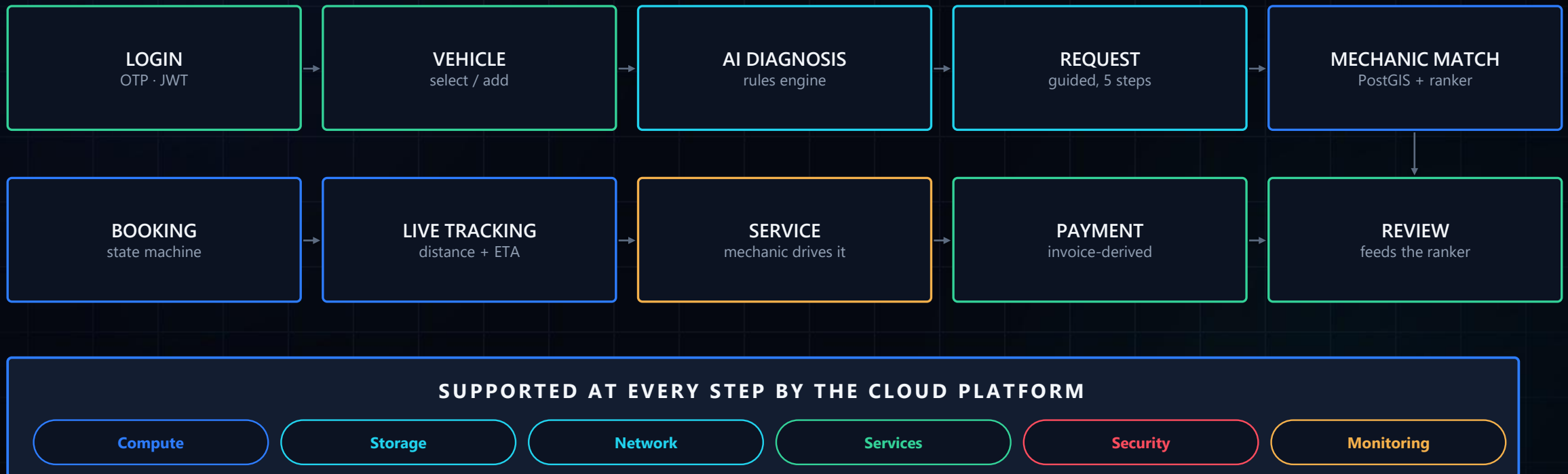
Closest real analogue in the project: dispatch is a scheduler. It observes which mechanics are available and where, then allocates a job — a dynamic scheduling decision at application level.

ROADASSIST BHARAT — CLOUD ARCHITECTURE



Right-hand badges give the honest status of each layer.

FROM BREAKDOWN TO RESOLUTION



Verified end to end by an automated browser suite driving this exact journey across both applications — 115 browser assertions, plus 165 API, 26 security and 22 payment-gateway assertions.

WHEN THE ROAD STOPS, ROADASSIST BHARAT KEEPS YOU MOVING.

AI + CLOUD + MOBILITY + SAFETY

AI

CLOUD

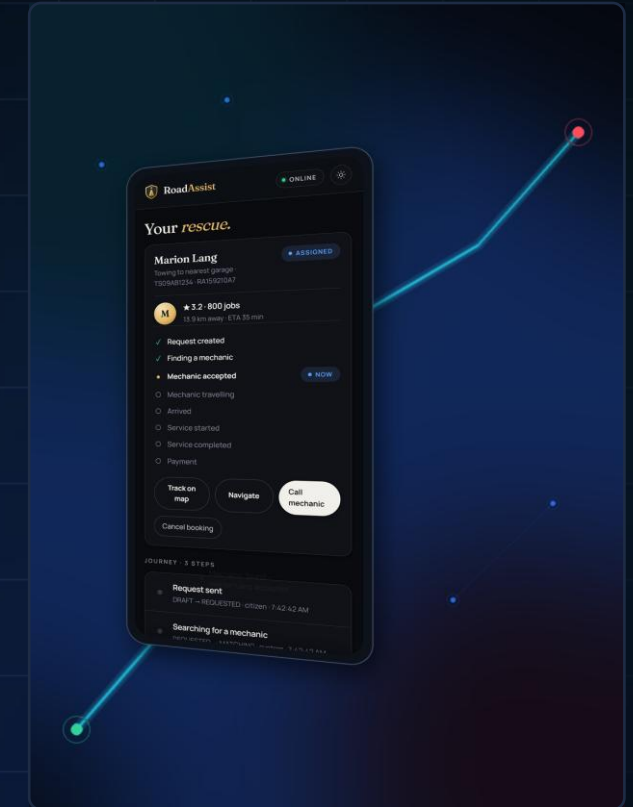
MAP

MECHANIC

RAKSHA

SAFETY

A working platform, used to demonstrate Cloud Computing Modules 1 through 4.



ACADEMIC MODULE MAPPING

MODULE 1

Slides 4, 5, 6, 7, 8

UNDERSTANDING CLOUD COMPUTING

Origins and influences · basic concepts and terminology · goals and benefits · risks and challenges · roles and boundaries · cloud characteristics · delivery models (IaaS/PaaS/SaaS) · deployment models

MODULE 2

Slides 9, 10, 11, 12, 13

CLOUD ENABLING TECHNOLOGY

Data center technology · virtualization technology · web technology · multitenant technology · service technology (Razorpay gateway integration)

MODULE 3

Slides 14, 15, 16, 17, 18

CLOUD INFRASTRUCTURE MECHANISMS

Network perimeter · virtual server · cloud storage device · cloud usage monitor · resource replication

MODULE 4

Slides 19, 20, 21, 22, 23, 24

FUNDAMENTAL CLOUD ARCHITECTURES

Workload distribution · resource pooling · dynamic scalability · elastic resource capacity · service load balancing · cloud bursting · elastic disk provisioning · redundant storage · migration · static scheduling · dynamic scheduling

VIVA PREPARATION

What is cloud computing?

On-demand network access to a shared pool of configurable computing resources, provisioned and released with minimal management effort. RoadAssist consumes compute, storage and network this way rather than owning hardware.

What are the characteristics of cloud computing?

On-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. Our PWA and booking flow demonstrate the first two directly.

Difference between public, private and hybrid cloud?

Public is shared provider infrastructure; private is dedicated to one organisation; hybrid combines both. Hybrid suits us: public for the citizen app, private for RAKSHA authority and medical data.

Why is virtualization useful?

Consolidation, isolation and portability. Several workloads share one machine safely, and a workload can move between machines because it no longer depends on specific hardware.

Why does RoadAssist need cloud computing?

Because the information a stranded driver needs is spread across parties and must be combined in real time, and because demand is uneven and geographically dispersed. One authoritative service reachable from any road is the requirement.

What are IaaS, PaaS and SaaS?

Infrastructure as a Service gives virtual machines and storage; Platform as a Service adds a managed runtime; Software as a Service delivers finished software. RoadAssist IS a SaaS application; IaaS and PaaS are how it would be hosted.

What is virtualization?

Abstracting physical hardware so one machine presents itself as several isolated logical ones. We use OS-level virtualization - Docker containers - not a hypervisor we operate.

What is multitenancy?

One running instance serving multiple independent tenants with isolated data. Our tenants would be vehicle fleets and government jurisdictions; the schema exists and queries are scoped to the caller.

VIVA PREPARATION (cont.)

What is a virtual server?

A software-defined server sharing physical hardware with others, allocated on demand and isolated from neighbours. Our API is stateless, so it could run on any number of them.

What is cloud storage?

Persistent, network-addressable storage provided as a service. Ours is PostgreSQL 16 with PostGIS - 61 tables including a hash-chained audit log.

What is cloud usage monitoring?

Collecting runtime usage and performance data for billing, capacity planning and scaling decisions. We have a limited form: a health endpoint reporting database latency, and per-request logging.

What is resource pooling?

Grouping resources to serve many consumers from a common pool assigned on demand. Our shared map-tile cache is a small working example.

What is a network perimeter?

The controlled boundary between untrusted networks and internal services. Ours layers TLS, rate limiting, schema validation, authentication, role gating and resource-level ownership checks.

What is resource replication?

Creating multiple instances of a resource so load can be spread and a single failure does not lose the service. Not implemented for us - we run a single database instance.

What is workload distribution?

Spreading incoming work across multiple resources so no single one is overwhelmed. Our five request types - AI, booking, map, polling, SOS - have very different shapes.

What is dynamic scalability?

Automatically adding or removing resource INSTANCES as demand changes - scaling out. Not implemented; the stateless design permits it.

VIVA PREPARATION (cont.)

What is elastic resource capacity?

Resizing existing resources - more CPU or memory to the same server - rather than adding instances. Scaling up rather than out.

What is cloud bursting?

Running baseline load privately and overflowing into public cloud during peaks, then releasing it. Our festival and monsoon peaks are exactly this shape.

What is redundant storage?

Duplicate copies on separate devices or sites so failure does not destroy data. Distinct from backup, which recovers an earlier point in time.

What is static scheduling?

Allocation decided in advance and held throughout execution. Predictable and cheap, but blind to actual runtime load.

What is load balancing?

Distributing requests across instances and health-checking them, so an unhealthy instance leaves rotation. It serves availability as well as throughput.

What is elastic disk provisioning?

Storage allocated as data arrives and billed on use, rather than pre-purchased at a fixed size. Not implemented - we use a fixed local volume.

What is migration?

Moving a workload or resource between hosts - for maintenance, consolidation, cost, or to escape a failing node.

How is the payment gateway integrated securely?

The amount comes from the invoice, never the request; the key secret never leaves the server; and settlement requires an HMAC-SHA256 signature the server re-computes. A webhook settles even if the payer's browser never returns.

VIVA PREPARATION (cont.)

What is dynamic scheduling?

Allocation recomputed during execution as demand changes. Our dispatch engine is a dynamic scheduler at application level: it ranks live mechanics by distance, rating and experience at request time.